

Liquidity Shocks in Virtual Markets

Caleb Bray

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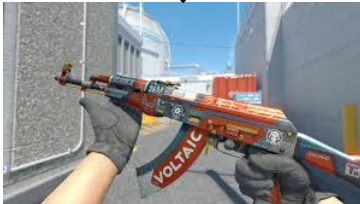
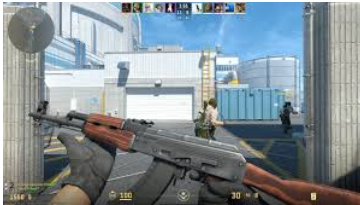
Introduction

- This paper serves as a case study to estimate the effects of sudden market structure changes in virtual asset markets.
- I use a DID specification to identify a 4.2% decrease in quality-adjusted prices and 34.4% drop in trade volume after implementation of a 7-day trade ban on a subset of items listed on a virtual exchange.
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Contribution

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Skins Overview



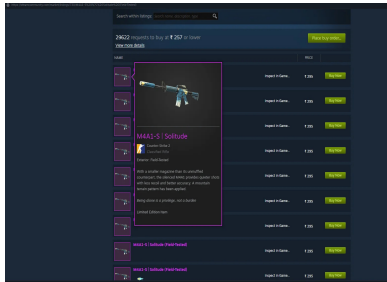
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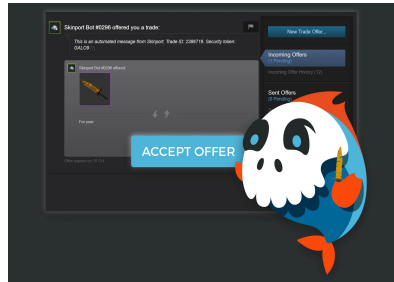


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The Marketplaces

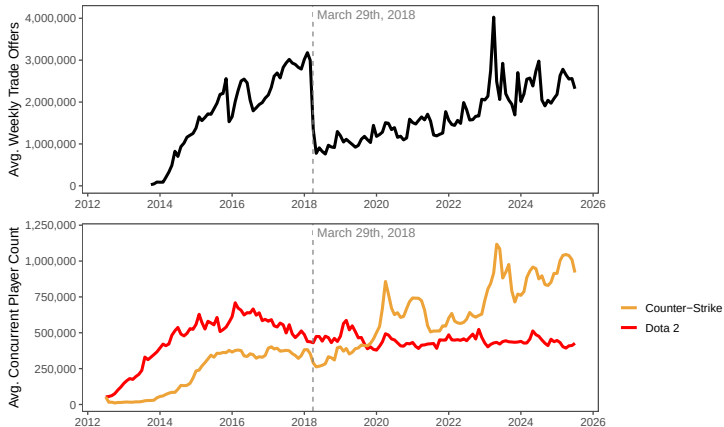


Steam Community Market Listings



Trade with a Third-Party Bot Account

The 3/28/18 Trade Ban



Note: Steam trade offer data includes trade offers for all Steam items and is estimated from the difference in trade offer ID between trades. Top chart is sourced from Trade.TF and bottom figure is sourced from Steam Charts.

Dataset

Counter-Strike and Dota 2 Items Summary

	Counter-Strike	Dota 2	Combined
Observations	2940	2675	5615
Unique Items	60	56	116
Oldest Origin Date	2013-08-14	2014-05-08	2013-08-14
Newest Origin Date	2016-02-18	2016-10-05	2016-10-05
Median Sale Price	\$4.26	\$1.90	\$4.13
	(\$7.28)	(\$3.61)	(\$7.14)
Log Median Sale Price	0.517	-0.064	0.484
	(1.379)	(1.160)	(1.253)
Volume Sold	20524.25	1356.84	11392.85
	(37685.37)	(2201.17)	(28938.78)
Grade Rarity	0.154	0.110	0.152
	(0.109)	(0.065)	(0.107)

Note: Median Price and Grade Rarity are monthly averages weighted by volume sold with weighted standard deviation in parentheses below. Volume Sold is a monthly average with unweighted standard errors. Origin date and grade rarity are provided by CS2 Database and Backpack.TF respectively.

Regression Results

Quality Adjustment and DID Regression Results

	$\log(p_{it})$		$\log(\hat{p}_{it})$	$\log(\text{Volume Sold})$
β_0	1.595*** (0.075)	$\log(\text{ConcurrentPlayers})$	-0.050*** (0.012)	-0.422*** (0.120)
<i>ItemAge</i>	0.029*** (0.003)	<i>Treated</i> $\times$ <i>Post</i>	-0.043*** (0.010)	-0.326*** (0.081)
<i>ItemAge</i> ²	-0.001*** (0.000)	N	5579	5615
<i>GradeRarity</i>	-19.43*** (0.894)	Adjusted R^2	0.997	0.956
<i>GradeRarity</i> ²	38.37*** (2.623)	Within R^2	0.130	0.055
N	5615	Item and Month FEs	✓	✓
Adjusted R^2	0.452			

Note: * $p < 0.5$; ** $p < 0.01$; *** $p < 0.001$. Standard errors are shown in parentheses below their respective estimates. Both regressions with price as a dependent variable are weighted by Volume Sold. Item Age is in terms of months.

Conclusion

